

Skin Lesion Triage

Final Presentation

Zohreh Asadi, Jannis Ole Röhl, Fenya Waßerroth, David Celik
Automatic Image Analysis, July 6, 2026

Recap - Skin Lesion Triage Project



Dataset HAM10000: dermoscopy images with diagnostic labels

Core task: Binary melanoma-vs-benign triage with calibrated uncertainty

Main question: How should a classifier behave when false negatives are much more costly than false positives?

Modern pipeline:

→ Frozen EfficientNet-B0 with a trained classification head

→ ☐ Uncertainty: Monte Carlo Dropout

Data split:

Subset used for training process		80%	Test set	20%
Train set	70%	Validation set	10%	

Class Imbalance: 83,21% benign & 16,79% malignant

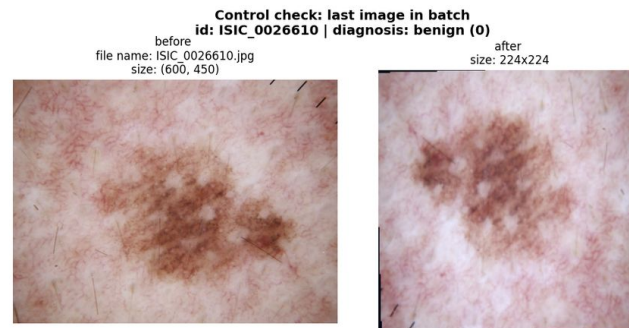
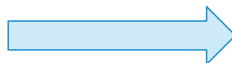
→ downsampled benign class in train set to avoid the classifier to only predict benign

→ balanced train set contains 2300 images

Specific image preprocessing for modern pipeline:

```
train_transforms = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.RandomHorizontalFlip(),
    transforms.RandomVerticalFlip(),
    transforms.RandomRotation(degrees=15),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])
```

```
test_transforms = transforms.Compose([
    transforms.Resize((224, 224)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
])
```



Modern Pipeline based on pre-trained EfficientNet-B0



Structure classification head:

```
model.classifier = nn.Sequential(
```

```
    nn.Dropout(p=dropout_rate),  
    nn.Linear(num_fts, 128),  
    nn.ReLU(),
```

} added another layer for catching
non-linearity in data

```
    nn.Dropout(p=dropout_rate),  
    nn.Linear(128, 1)
```

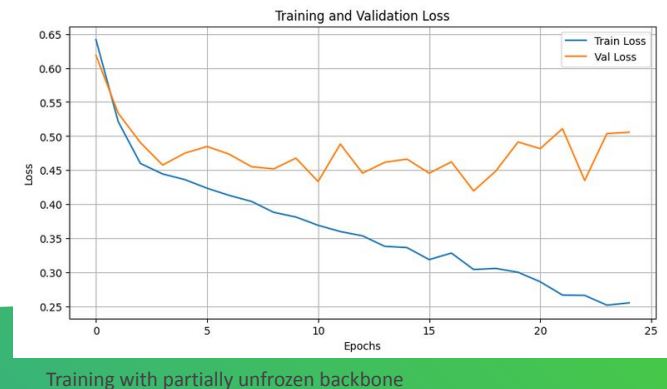
} started with a very simple
classification head

Parameter fine-tuning:

- **learning rate:** 0.001
- **epochs:** 40
- added **early stopping** after 7 epochs without further improvement

Experimented with partially unfreeze backbone:

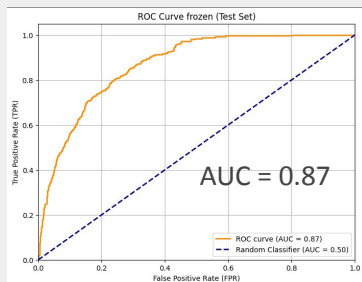
- last two layers
- smaller learning rate (0.0001) to avoid forgetting fundamental learning from pre-training



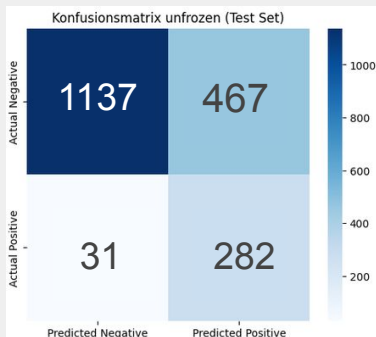
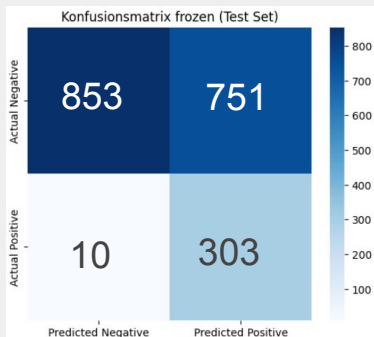
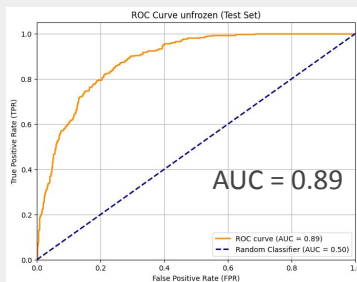
Results from testing modern pipeline



frozen



unfrozen



Threshold = 0.3

Recall **frozen 96.81%**

Precision **28.75%**

Accuracy **60.30%**

unfrozen

90.10%

37.65%

74.02%

Setting Threshold:

- analyzed various **thresholds** based on the validation set
→ higher threshold yields higher precision but lower recall

Threshold	Precision		Recall	
0.1	22.77%	31.07%	99.39%	97.56%
0.2	26.09%	35.68%	98.17%	95.73%
0.3	29.02%	39.36%	95.73%	90.24%
0.4	33.55%	43.33%	93.90%	87.20%
0.5	40.11%	46.55%	85.37%	78.05%

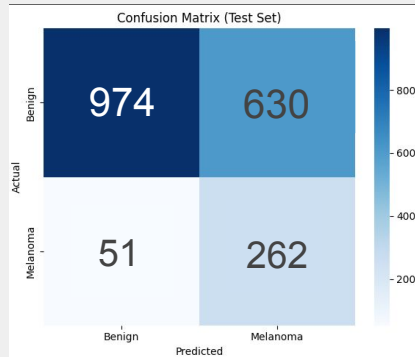
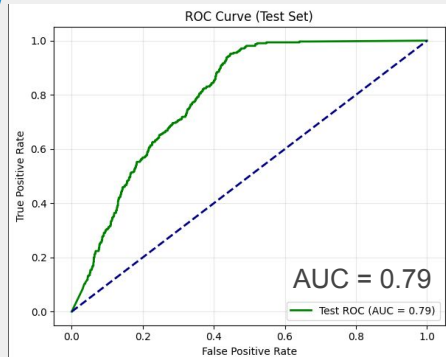
Impact of partially unfreeze backbone:

- unfreezing layers 7 & 8 gave the model more capacity
→ the model adapted more strongly to the majority class
- the unfrozen model achieves a higher global accuracy by classifying a larger portion of ambiguous cases as "negative"
- this shift improves precision but lowers recall
→ frozen: 10 FN
→ unfrozen: 31 FN

Comparison classical pipeline vs. modern pipeline



classical pipeline

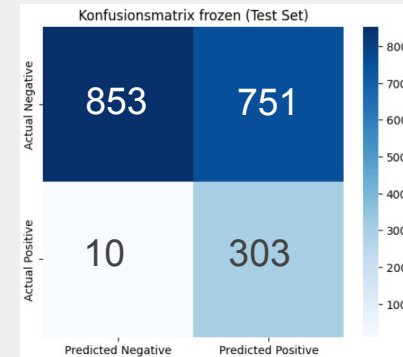
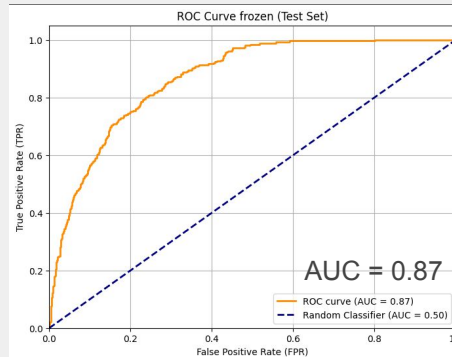


Precision
29.37%

Recall
83.71%

Accuracy
64,00%

modern pipeline



Precision
28.75%

Recall
96.81%

Accuracy
60,30%

When evaluated on the identical test set of 1,917 images, the **modern pipeline proves significantly safer for clinical triage** than the classical GMM approach. Not only does it increase the AUC from 0.79 to 0.87, but it also **reduces the risk of missing a malignant skin lesion** by over 80%. Specifically, the miss rate drops from 16.3% to just 3.2%, leaving only 10 False Negatives compared to the 51 missed by the classical model.

Thanks for your attention!
Questions?

Skin Lesion Triage

Group 42

Projects - Presentation 1

Soliman Andrawes
Anja Capo
David Martin Obina Gottschalk
Tomoki Kai Mengel

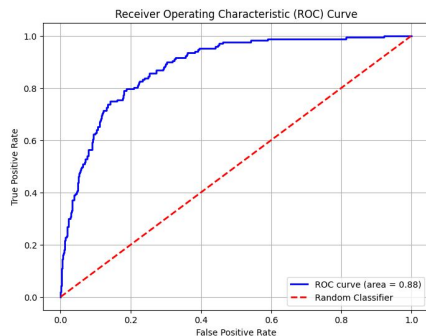
Dataset Split

- data split into 70 (train), 15 (validation), 15 (test)
- class stratification to ensure even distribution of different image types (mel vs benign) over the 3 datasets
- computation of hsv histograms and writing to file → cached data

```
train images= 6978 mel= 783 (11.22%)  
val   images= 1533 mel= 164 (10.70%)  
test  images= 1504 mel= 166 (11.04%)
```

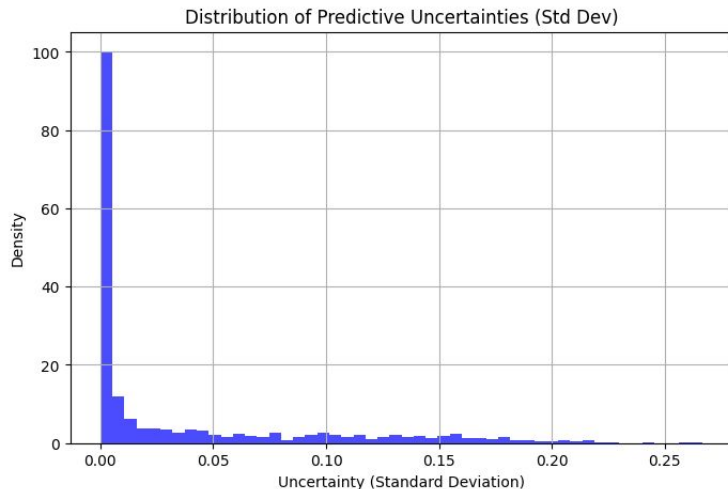
Modern Pipeline

- Frozen EfficientNet-B0
- classification head with dropout layers
- Disable gradient updates to “freeze” the backbone
- Enable the dropout layers for the uncertainty estimation by toggling training mode during inference
- DataLoader



MC-Dropout uncertainty estimation

- We take samples
- Std as uncertainty estimation of the predictions



Bayesian Decision Theory Layer

The classical model first trains Gaussian Mixture Model for each class:

$$P(x | \text{benign}), P(x | \text{melanoma})$$

Using Bayes' rule, we convert the GMM likelihoods into posterior probabilities:

$$P(\text{class} | x) \propto p(x | \text{class}) * P(\text{class})$$

Then we apply Bayesian decision theory with asymmetric loss.

False negative:

Actual melanoma, predicted benign $\rightarrow$ high cost

False positive:

Actual benign, predicted melanoma $\rightarrow$ lower cost

We define:

$$C_{\text{FN}}=10, C_{\text{FP}}=1$$

Expected risks:

$$R(\text{predict melanoma} | x) = C_{\text{FP}} \cdot P(\text{benign} | x)$$

$$R(\text{predict benign} | x) = C_{\text{FN}} \cdot P(\text{melanoma} | x)$$

Final decision:

Choose the prediction with the lower expected risk

Classical Pipeline: Parameter Optimization

Goal: Select the best Gaussian Mixture Model (GMM) parameters for melanoma triage.

- Train one **Gaussian Mixture Model (GMM)** per class
 - benign
 - melanoma
- Hyperparameters tested:
 - Number of components: **1, 2, 4, 8**
 - Covariance type: **diag, full**
- Validation metric:
 - **Asymmetric Bayesian decision cost**
 - not standard accuracy
- Cost setup:
 - False Positive cost = **1**
 - False Negative cost = **10**

Best parameters are selected using the validation set, prioritizing melanoma detection over overall accuracy.

Cost-Sensitive Evaluation & Pipeline Comparison

Classical pipeline

Uses Bayesian decision threshold:

$$T = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{1}{11} = 0.091$$

- classify as melanoma already at **9.1% probability**
- reduces dangerous false negatives
- Fast and interpretable
- Limited feature representation

Modern pipeline

Evaluated using:

- Standard threshold: **0.50**
- Cost-sensitive threshold: **0.091**

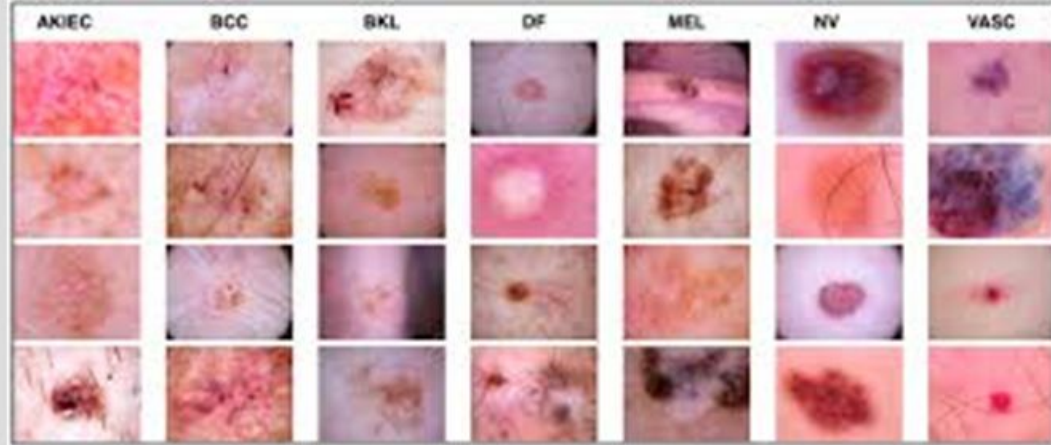
This allows direct comparison with the classical model.

- Richer learned features
- Better predictive performance

Automatic Image Analysis

Topic B: Bayesian Decision and Uncertainty: Skin Lesion Triage

Presentation 2: Modern pipeline & Comparison



Group 20:

Malte Jonas Kriese
Yannik Queisler
Furkan Akgün

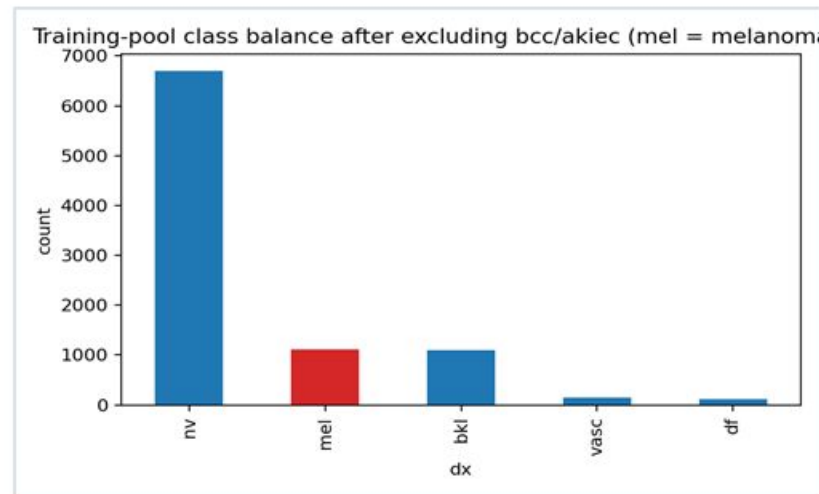
RECAP

Classical Baseline & the Core Question

Classical Approach

- Convert dermoscopy images to HSV, extract per-channel histograms (32 bins)
- Fit a Gaussian Mixture Model per class (K via BIC: 28 benign, 8 melanoma)
- Apply Bayes' rule for posteriors, then a MAP vs. cost-sensitive decision

Main question: how should the classifier behave when missing a melanoma is far more costly than a false alarm?



Training-pool class balance (melanoma in red)

1,860

held-out test images

10.8%

melanoma prevalence

0.143

classical ECE

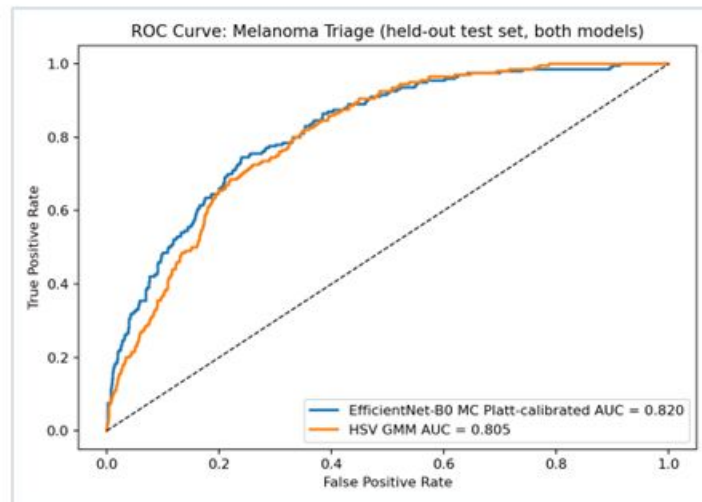
54%

melanomas <1% risk

EfficientNet-B0 with a Fine-Tuned Head

Architecture & Training

- ImageNet-pretrained EfficientNet-B0 backbone + new classification head
- Stage 1: freeze backbone, train head only (6 epochs)
- Stage 2: unfreeze last 2 blocks, fine-tune at low LR (6 epochs)



ROC curve, held-out test set ($n = 1,860$)

0.820

EfficientNet test AUC

0.805

classical GMM AUC

+0.015

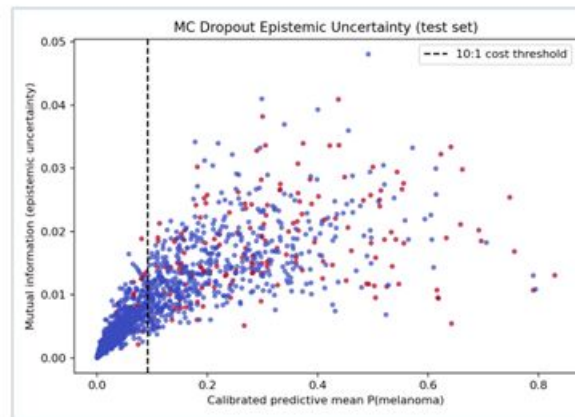
AUC improvement

UNCERTAINTY

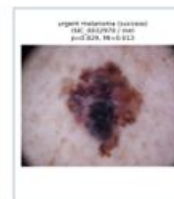
Knowing What the Model Doesn't Know

How It Works

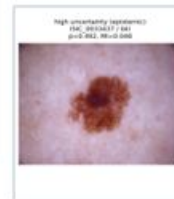
- Keep dropout active at inference; run 30 stochastic forward passes
- Averaging the passes approximates a Bayesian posterior over weights
- Predictive entropy splits into aleatoric (data noise) + epistemic (model unsure)
- High epistemic uncertainty -> route to manual review, not an automatic call



Mean vs. epistemic uncertainty



Confident (low MI)



Uncertain (high MI)

21.2%
auto-reassured

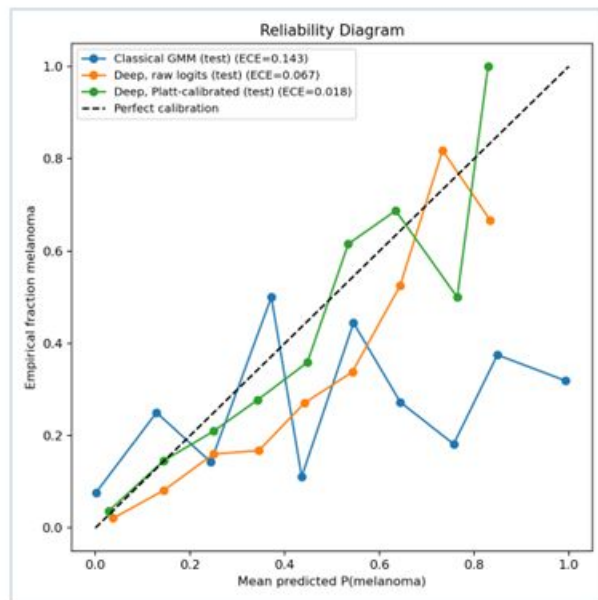
76.2%
manual review

2.6%
urgent referral

98.5%
sensitivity kept

COMPARISON

Whose Confidence Can You Trust?



Closer to the diagonal = better calibrated

Calibration error (ECE, lower is better)

0.143

Classical GMM

0.067

Deep, raw logits

0.018

Deep, calibrated

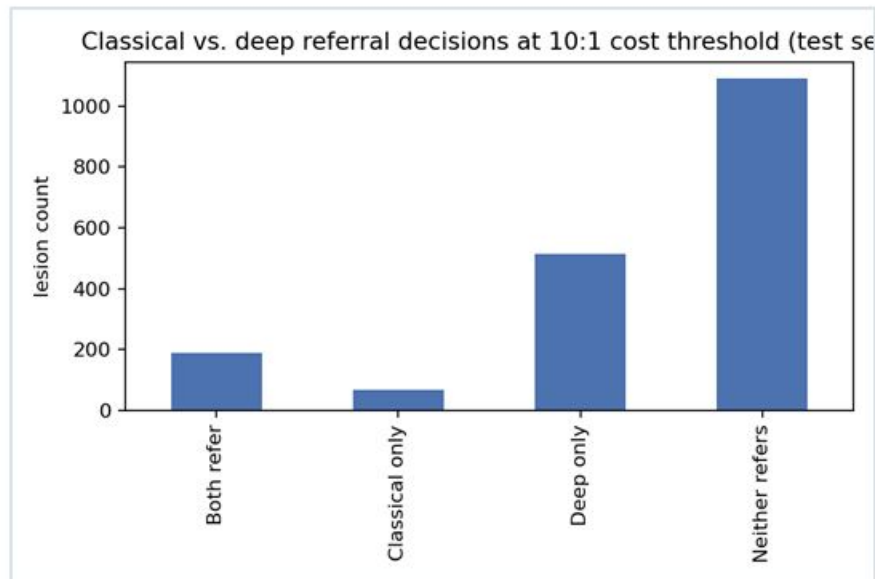
Reaching safety-critical sensitivity (95%+)

Model	Sensitivity	Referral needed
Classical GMM	95%+	100% (all lesions)
EfficientNet-B0	97.5%	68%

Classical GMM's posteriors don't separate classes well enough for safety-critical use; EfficientNet does.

COMPARISON

Classical vs. Deep: Which Model Wins?



Referral decisions at the 10:1 cost threshold

68.8%

agreement
between models

20/200

melanomas missed
by both

514

images deep
model alone flags

Recommended: EfficientNet-B0 + cost-sensitive threshold ($\tau = 0.091$) + uncertainty-aware triage.

Topic B: Bayesian Decision and Uncertainty: Skin Lesion Triage

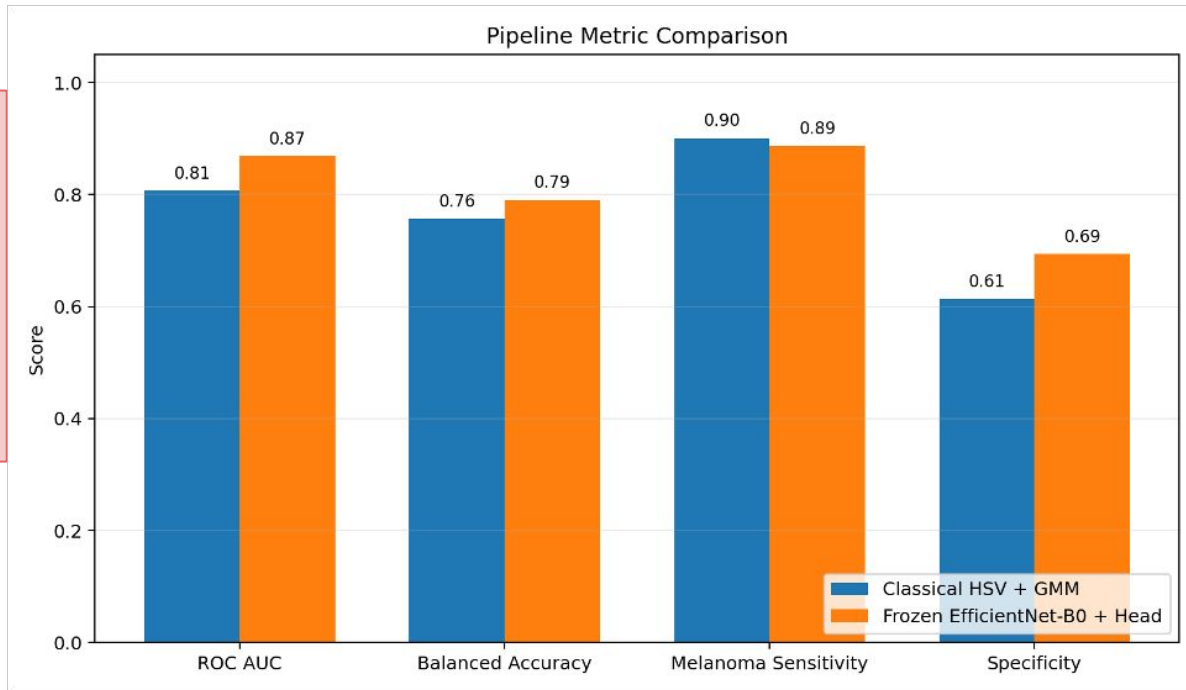
Performance Comparison

EfficientNet-B0 improves discrimination and reduces false alarms.

The trade-off is clinical: it misses 3 additional melanoma cases compared with HSV + GMM.

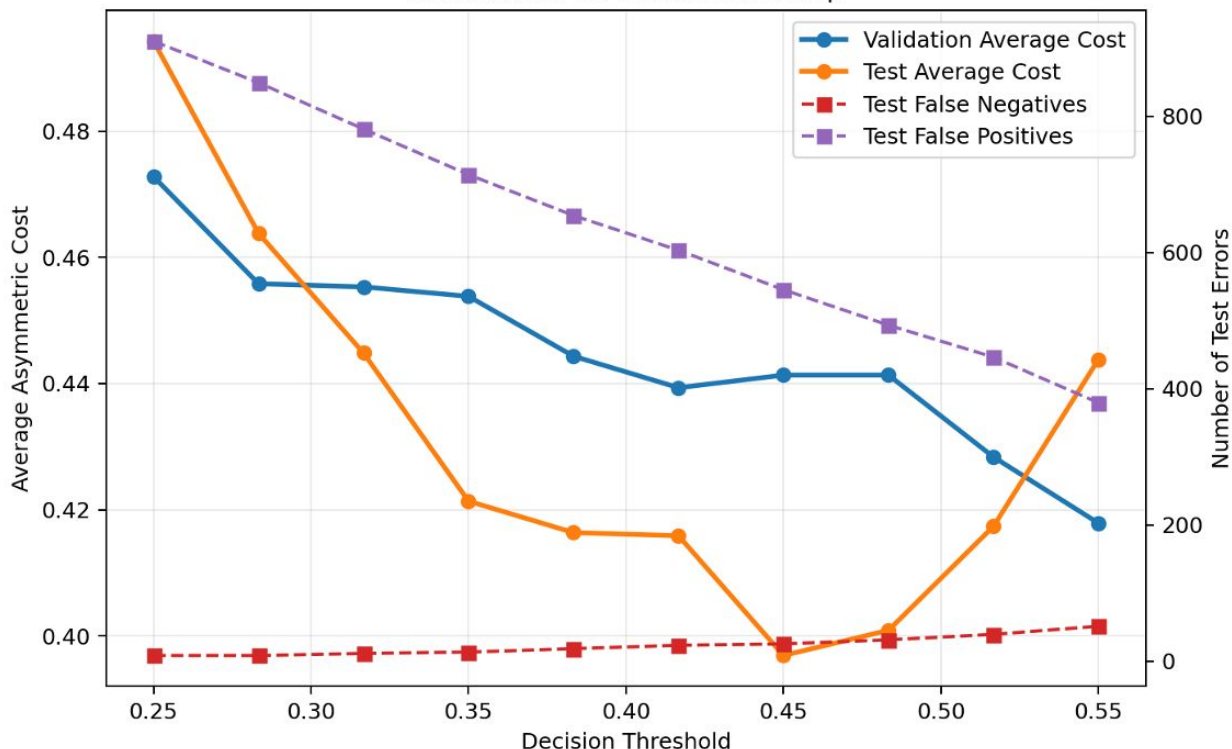
Group: 17

Natalie Seiffert
Finn Frederick Hans Watermann
Maximilian Johannes Winkler
Ekaterina Zhittsova



Threshold Optimization

EfficientNet-B0 Threshold Sweep



Default 0.50 is arbitrary. It assumes symmetric errors and calibrated probabilities.

Medical triage is asymmetric. FN is clinically more dangerous than FP, so the operating point must reflect cost.

Observed optimum: test average cost reaches minimum around threshold 0.45.

Caveat: final threshold should be selected on validation data, then reported once on test.

Clinical Cost Evaluation

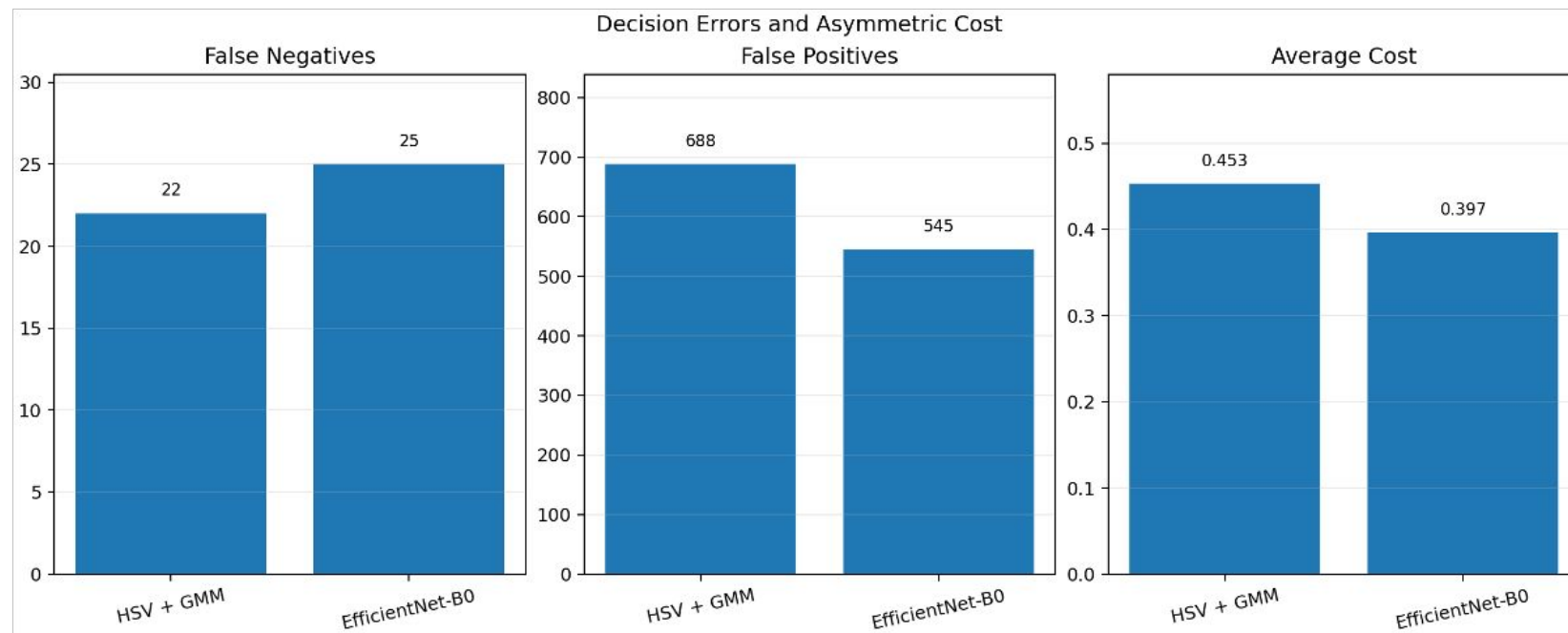
Cost-Sensitive Result

Average cost: 0.453 $\rightarrow$ 0.397; relative reduction $\approx 12.4\%$.

Operational value: lower FP count can reduce unnecessary follow-up and biopsy pressure.

Safety constraint: FN must remain bounded because missed melanoma is the clinically costly error.

Deployment view: model should be a triage aid, not a standalone diagnosis.



Final Conclusions

Key Takeaways

- 1) Better overall discrimination. EfficientNet-B0 improves AUC from 0.807 to 0.868.
- 2) Better operating-point economics. Average cost drops from 0.453 to 0.397.
- 3) Main clinical trade-off. False positives decrease strongly, but false negatives rise from 22 to 25.

EfficientNet-B0 improves ranking and reduces clinical workload, but threshold choice determines whether the model is medically acceptable

Pipeline	AUC	Bal. Acc.	Sens.	FN	FP	Avg. Cost	Threshold
HSV + GMM	0.807	0.757	0.901	22	688	0.453	-2.755
EffNet-B0 + Head	0.868	0.791	0.887	25	545	0.397	0.450

